

Session 4 Solutions

Lecture 41

1. *Explain the term message digest? Explain why a 64 bit message digest is vulnerable to misuse.*

A small but unique summary of a document. A 64 bit Message Digest is vulnerable to a birthday attack whereby two different documents can be constructed with the same Message Digest.

2. *Fasthash is a special purpose chip that can calculate the message digest of a standard legal document in 10^{-6} seconds. The message digest produced is a convenient 32 bits and can be stored as an integer. Lawyer Bob has prepared 2 different documents entitled “Rip Off” and “Sweet Deal” respectively. He knows the message digest produced by the “Sweet Deal” document. Describe how he might modify the “Rip Off” document so that it produces the same message digest as the “Sweet Deal” document. Roughly how long will it take him to do so, assuming that document editing time is negligible.*

Describe in detail how he could achieve the same aim in less time.

There are 2^{32} different message digest values. If lawyer Bob produces 2^{32} different legal documents some will have identical message digest, but he has a good chance of finding one which will match the Sweet Deal document. He can do this by making invisible changes in 32 different places. 2^{32} is approximately $4 * 10^9$ and he can check 10^6 documents in a second, so it will take $4 * 10^3$ seconds, which is 4000. This is just over an hour.

If he edits both the Sweet Deal and the Rip Off documents then he can find a match faster using a table lookup algorithm. This is a birthday attack. If he makes 2^{16} different version of Sweet Real and adds them to a table then he will need to make about 2^{16} different versions of the rip off document to find a match. 2^{16} is roughly 64,000, so calculating the hashes would take about 1/8 of a second (2 lots of 64k).

Lecture 42

3. *Describe the differences between real random number, pseudo random number and cryptographically secure pseudo random number sequences. Give one algorithm for producing each of the above types of random numbers (three algorithms in total).*

A real random number sequence cannot be predicted. A pseudo-random number sequence has the same distribution as real random numbers but can be generated

again if required. It will have a seed to start things off and using the same seed generates the same sequence. Normally it is possible to predict the next number if the previous numbers are known, but this is not possible with a cryptographically secure pseudo-random number sequence.

Keyboard latency will provide truly random numbers, with a few bits per keystroke. A linear congruential generator $x_n = a * x_{n-1} + b \pmod{c}$ will provide a good pseudo-random sequence provided the values of a , b and c are chosen well. Elliptic curves can generate cryptographically secure pseudo-random numbers.

Lecture 43

4. *An organisation has decided that its secrets are too valuable to entrust to just one person, and has decided that three people will be needed to access the secret information. Devise a scheme that will allow the head of the organisation to issue passwords to three people in such a way that all three passwords are needed simultaneously. Give a numerical example of the operation of your scheme when the secret is the number 42. You may assume that each secret is a 6 bit number.*

Pick two random bitstrings r_1 and r_2 and hand out r_1 , $s \oplus r_2$, $r_1 \oplus r_2$ as the three keys ($\oplus$ is the exclusive or operator). The secret can be recovered by exclusive oring the three keys. The numerical example converts 42 to a bitstring 101010, choosing two random bitstrings, such as 110011 and 011101 and performing the above calculations to generate the keys 110011, 110111, 101110. Combining these three results in $1 \oplus 1 \oplus 1 = 1$ for the first bit, and so on, recovering 101010.

5. *One drawback of the previous scheme is that all three persons are needed to operate it. All secrets would be lost if one person were to have an accident. Devise an alternative scheme where three people are still needed to access the information, but five people have parts of the key. Any three people will be sufficient. A numerical example is not needed!*

A Lagrangian interpolation scheme would work for this case. The basic curve is $y = ax^2 + bx + s \pmod{n}$, where a and b are random and n large enough. 5 values of x are chosen at random, and the resulting values of y (together with x) handed out as keys. Any three of the (x, y) pairs can be used to recover the polynomial and hence find s .